Faysal Mahmud

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Professional Summary

Computer Science and Engineering undergraduate at AIUB building full-stack web applications with React, TypeScript, Python (FastAPI), PHP, and MySQL, backed by automated tests, CI, and Docker. Co-author of two machine learning papers on explainable intrusion and phishing detection, presented at ICACT, Nepal. Motivated to build reliable, maintainable software in an international engineering team.

Education

American International University-Bangladesh (AIUB) **Expected September 2027**
Bachelor of Science in Computer Science and Engineering *Dhaka, Bangladesh*

Relevant Coursework

- Advanced Programming in Python
- Object-Oriented Programming
- Data Structures & Algorithms
- Software Engineering
- Database Systems
- Operating Systems
- Web Technologies
- Artificial Intelligence

Technical Projects

ResearchFlow — Research Management Workspace | *React, TypeScript, FastAPI, Docker* GitHub **September 2026**
- Built a full-stack research workspace: LaTeX editor with version history, BibTeX manager, and experiment tracking.
- Designed a versioned FastAPI REST API with server-enforced roles, Argon2 password hashing, and rate limiting.
- Wrote 43 pytest and 22 Vitest tests; GitHub Actions CI runs Ruff, strict mypy, pip-audit, and Gitleaks; API ships in Docker.

FlowSense — AI Cash-Flow Copilot | *Next.js, TypeScript, FastAPI, PostgreSQL* Live **September 2026**
- Co-built in a team of four a copilot that plans how small businesses cover cash gaps; I built the frontend.
- Safety kernel enforces single-use approvals, exactly-once payments, and a hash-chained audit log; 261 tests run in CI.

Online Medicine Shop | *PHP, MySQL, JavaScript (AJAX), HTML5, CSS3* GitHub **May 2026**
- Built registration, login, profile, and browsing features for a three-member team web shop, merged via pull requests.
- Secured flows with salted password hashing, parameterized SQL, and CSRF tokens; added an AJAX JSON search endpoint.

Research & Publications

Machine Learning Research **Since 2025**
Undergraduate Researcher, Department of Computer Science, AIUB *Dhaka, Bangladesh*
- EV charging: removed 10 leakage-prone features, used train-only SMOTE, and compared four models (Macro F1 0.9748).
- Phishing detection: compared ML models on 1,082,664 URLs from eight public sources, with SHAP explainability.

Publications

1. N. I. Rishad, **F. Mahmud**, I. I. Orpa, R. A. M. Rudro, and K. Nur, "Explainable and Leakage-Free Intrusion Detection for EV Charging Station Authentication Attacks," *ICACT, Nepal*.
2. T. A. Fahim, **F. Mahmud**, S. J. Snegdha, R. A. M. Rudro, T. Sayeed, and K. Nur, "Phishing URL Detection Using Machine Learning: A Comparative Study with SHAP Based Explainability on a Large-Scale Multi-Source Dataset," *ICACT, Nepal*.

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C++, C#, PHP, SQL
Frontend: React, TypeScript, Vite, Tailwind CSS, Three.js, HTML5, CSS3, AJAX
Backend & APIs: FastAPI, Pydantic, REST API design, PHP, authentication and role-based access control, CSRF protection
Databases: MySQL, SQLite, Microsoft SQL Server, relational schema design, migrations, parameterized queries
Testing & DevOps: pytest, Vitest, GitHub Actions CI, Docker, Docker Compose, Ruff, mypy, Git, GitHub, Linux/Unix
Data & ML: Random Forest, XGBoost, LightGBM, SHAP, SMOTE, leakage-aware evaluation, LaTeX/Overleaf

Leadership & Achievements

AIUB Research & Development Club **Since January 2025**
Organizer *Dhaka, Bangladesh*
- Helped organize research seminars, methodology workshops, and events on machine learning and Erasmus Mundus.
- Co-organized seminars with Prof. Dr. Nowshad Amin (≈150 attendees) and Prof. Pushp Lata (260+ registrants).

Achievements

- **Winner**, Abstract Presentation Competition, State University
- **Winner**, University Physics Poster Presentation Competition
- Second round, Bangladesh / GP FutureMakers; Participant, CUET Three Minute Thesis (3MT) and Sciblitz 2.0 (CUET)